

Coarsening at Random

A small monograph on masked-data identifiability across domains: the one idea, its six instances, and the papers that are still missing

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1 What this is, and why it exists

The **coarsening-*** repositories are a family of papers that all turn out to be the same paper wearing different clothes. This document states the shared idea once, at the generality that explains the recurrence, places each existing paper inside that frame, and then does the thing a scattered family of preprints cannot do for itself: it draws the boundary of the program and names the papers that are *missing*, the gaps that, once filled, would close the framework.

It is written for two readers. The first is someone meeting the program for the first time, who wants the whole arc in one sitting rather than ten preprints. The second is the author, deciding what to build next, and whether the program is better told as a swarm of short papers or as a few deep ones. (The honest answer, developed in the coda, is the latter; this monograph is a first draft of the spine such a consolidation would need.)

The existing pieces, by Zenodo concept DOI:

Role	Paper	Concept DOI
Foundation	masked-causes-in-series-systems	10.5281/zenodo.18725577
Pillar 1 (C2 holds)	coarsening-synthesis	10.5281/zenodo.20533912
Pillar 2 (C2 fails)	coarsening-sensitivity	10.5281/zenodo.20604314
C2 companion (reliability)	mdrelax	10.5281/zenodo.20414727
Application	scrna-coarsening	10.5281/zenodo.20414734
Application	spatial-coarsening	10.5281/zenodo.20422883
Application	dp-coarsening	10.5281/zenodo.20422885
Application	weaksup-coarsening	10.5281/zenodo.20422888
Application	phenotype-coarsening	10.5281/zenodo.20422890
Application	mil-coarsening	10.5281/zenodo.20502964

2 The one idea

2.1 The shape every problem shares

You rarely observe the quantity you care about. You observe a *coarsening* of it: a report R that is a many-to-one or noised function of a latent Y , produced by a mechanism you do not fully control, and you want a latent parameter θ that governs the law of Y . Write the latent model $g(y; \theta)$ and the coarsening channel $\kappa(r | y)$. The data law factors,

$$p(R | \theta) = \underbrace{g(Y; \theta)}_{\text{latent model}} \times \underbrace{\kappa(R | Y)}_{\text{coarsening channel}},$$

and the second factor is the nuisance that blocks naive likelihood inference. The report defines a *candidate set* $c(R) = \{y : \kappa(R | y) > 0\}$, the latent values it does not exclude: a finite set of suspects in the discrete case, a kernel-weighted continuum in the additive-noise case.

2.2 The three conditions

Coarsening at random (CAR; Heitjan and Rubin 1991, Gill, van der Laan and Robins 1997) is three conditions under which the nuisance becomes *eliminable*, so that maximizing the latent model restricted to the candidate set, the “face-value” likelihood, is legitimate:

- **C1 (support).** The truth is always inside the reported candidate set.
- **C2 (symmetry).** Within a candidate set the mechanism does not depend on which admissible value is the true one. This is the coarsened-data form of “missing at random,” and it is the fragile one.
- **C3 (parameter independence).** The mechanism does not depend on θ .

2.3 The three results that recur

Under CAR, three statements recur in every domain. The contribution of the program is to prove them once and recover each field’s named version as a corollary.

1. **A consistency identity.** At an interior maximum of the face-value likelihood, the model’s fitted mean of a *coarsening-sufficient statistic* T equals its empirical mean, $\mathbb{E}_{\hat{\theta}}[T(R)] = \bar{T}$. This is exponential-family moment matching in disguise; it appears under six names (cell-total, spot-level, release, agreement, code-frequency, bag-prevalence consistency). Its moral is uniform and counterintuitive: marginal-fit agreement is not evidence of unbiasedness.
2. **A rank condition.** The latent parameter is identifiable if and only if an *augmented candidate-set matrix* $\tilde{C}$ built from the reports has full column rank. Confounded directions (latent components never separated by any report) make it fail.
3. **A restoration device: the singleton.** When the rank condition fails, a report that pins the latent value to a point, a *singleton* candidate set $|c(R)| = 1$, restores column rank and hence identifiability. Every field already had its singleton and had not noticed it was the same object.

3 The two pillars and the foundation

3.1 Foundation: masked causes in series systems

The framework was born in reliability. A series system fails when its first component fails; the cause is masked, but a candidate set of suspects is observed. `masked-causes-in-series-systems` proves the C1-C2-C3 likelihood collapse distribution-agnostically and the augmented-candidate-set rank theorem. Everything downstream is this paper ported. `mdrelax` is its C2-sensitivity companion (an R package plus paper) that quantifies what happens to reliability estimates when the symmetry condition is relaxed.

3.2 Pillar 1: the C2-holds synthesis

`coarsening-synthesis` is the keystone. It states the consistency identity once, in two regimes: a regular exponential family, where the identity is an exact finite-sample equality; and a location family, where it is exact at a single report and a population first-moment identity otherwise, sharp only for the Gaussian kernel. It states the rank condition and the singleton restoration once. It then recovers six named domain theorems as corollaries, and it is candid about the two seams where the reduction is not clean: differential privacy lives in the location-family branch, not the exponential-family one; weak supervision reduces exactly only under a sufficiency-complete parametrization, asymptotically otherwise.

3.3 Pillar 2: the C2-fails sensitivity theory

coarsening-sensitivity develops the case the synthesis assumes away. It parametrizes the C2 violation by a *tilt of magnitude* δ and proves two things. First, a **sensitivity bound**: the asymptotic bias of the face-value MLE is, to leading order, linear in δ with an explicit constant, the inverse face-value information times the covariance of the score with the tilt, $\hat{\theta} - \theta^* \approx \delta \mathcal{I}^{-1} \text{Cov}(s, h)$, so the latent parameter is partially identified over a set that contracts to a point as $\delta \rightarrow 0$. Second, a **restoration sample complexity**: recovering the r confounded directions to a target accuracy takes of order r/γ^2 singletons, where γ is the domain’s identification margin. Pillars 1 and 2 are complementary: one says what is recoverable when coarsening is ignorable and how to restore it; the other says how wrong the fit is when it is not, and how much external data buys identification back.

4 Six fields, one structure

Each application instantiates the framework by naming four objects: the latent quantity, the coarsened report, the form a C2 violation takes (the tilt), and the singleton that restores identifiability. The reliability foundation is the zeroth row; the six below are the ports.

Domain	Latent Y	Coarsened report R	C2 violation (tilt)	Singleton
Reliability (foundation)	which component failed	candidate set of suspects	group-structured informative masking	a resolving diagnostic
scRNA-seq	true transcript count	zero-inflated observed count	dropout increasing in true expression	ERCC spike-in (known count)
Spatial tran- scriptomics	spot cell-type composition	pooled spot expression	capture efficiency varying by cell type	single-cell-resolution probe
Differential privacy	confidential statistic	privatized (noised) release	data-dependent mechanism (SVT, PTR)	a non-private release
Weak supervision	true label	vector of labeling-function votes	labeling-function dependence	a gold-labeled example
EHR phenotyping	true clinical state	set of diagnosis codes	severity-correlated coding	a chart-reviewed patient
Multiple instance learning	instance labels	bag (OR) label	within-bag instance dependence, firing rate	a singleton bag (instance label)

Read as mathematics the table is one theorem, one rank condition, one restoration proposition. Read as practice it is a catalogue of which auxiliary experiment each field should run and how much of it: the r/γ^2 rate of Pillar 2 turns “collect some gold labels, spike-ins, or chart reviews” into a budget with a domain-specific margin.

The instances are not equally clean, and the program says so. scRNA-seq, spatial, phenotyping, and MIL sit in the exponential-family regime where the consistency identity is exact. Differential privacy sits in the location-family branch (its “singleton,” a non-private release, is the device that

defeats privacy, so it is a thought experiment about identifiability, not a usable remedy). Weak supervision is exact only under a parametrization practitioners do not use.

5 The gaps: papers that are missing

This is the part a scattered family cannot write about itself. The program as it stands answers “when is the latent parameter identifiable, what is recoverable, and how does a singleton restore it.” It leaves whole questions untouched. Each gap below is stated as a paper that could be written, with the result it would establish and its relation to what exists. They are ordered by how much they would strengthen the spine.

5.1 Tier 1: structural gaps that change the foundation

G1. The semiparametric-efficiency pillar (the biggest gap). The entire program estimates by parametric face-value MLE. But CAR *is* the Gill, van der Laan and Robins setting, the home of semiparametric efficiency and targeted learning. The missing paper develops the influence-function, efficient-score and targeted-minimum-loss (TMLE-style) doubly-robust estimator for the coarsening estimand. Payoff: efficiency and robustness the current MLE lacks; a single semiparametric account that *dissolves* the awkward “regime A versus regime B” split of the synthesis (both become one efficient-influence-function statement); and the right language for nuisance estimation when the coarsening channel must itself be modeled. This is the paper that would move the program from “a recurring identity” to “the efficient theory of coarsened-data estimation.” High difficulty, highest value.

G2. Estimating the identifiability geometry (r, γ) from data. Pillars 1 and 2 *assume* the confounded dimension r and the margin γ are known. They are not: in practice the candidate-set geometry is itself estimated. The weak-supervision paper already flags “estimating the degeneracy dimension r from data” as open; it is open everywhere. The missing paper turns the rank condition and the margin into *estimable quantities* with their own sampling theory: a consistent estimator of the rank deficit, a confidence statement on γ , and a diagnostic a practitioner runs before trusting any instance-level or latent output. Without this, the r/γ^2 budget is not operational. Medium-high difficulty, very high value (it is what makes the whole program usable rather than diagnostic).

G3. Optimal and active design of singletons. The program identifies the singleton as the universal remedy but never optimizes its allocation. The r/γ^2 rate is a given-the-design statement; the missing paper asks the design question: which singletons to collect, in what proportion, to restore the confounded subspace fastest. This is optimal experimental design and active learning along the deficient directions, with the per-domain cost structure (a chart review is expensive, a spike-in is cheap) entering the objective. It converts the singleton from “a device that works” into “a budget you optimize.” Medium difficulty, high value, and unusually practical.

5.2 Tier 2: natural extensions of the existing pillars

G4. Global (non-local) sensitivity and sharp partial-identification regions. Pillar 2’s bias bound is first-order in δ : local. For large C2 violations it says nothing. The missing paper

computes *sharp* partial-identification regions (Manski or Tamer style) or higher-order expansions, so the partial-identification set is exact rather than a first-order ellipse. This is “Pillar 2 at finite δ .”

G5. Bayesian coarsening and Bayesian sensitivity. Everything is frequentist MLE. The missing paper puts a prior on the tilt instead of a worst-case bound (Daniels and Hogan spirit), yielding a posterior over the partial-identification region, and develops the Bayesian analogue of the consistency identity (posterior moment matching) and of the singleton (informative auxiliary likelihood). It also gives the program a natural account of *how much* a practitioner believes C2.

G6. Nonparametric and neural face-value models. The clean results lean on exponential or location families. The missing paper replaces the parametric face-value model with a neural density estimator or normalizing flow, enforcing the consistency identity as a moment constraint during training, and asks when identifiability survives. This is the bridge to deep generative scRNA-seq models (scVI, DCA) that currently handle dropout without identifiability guarantees; the coarsening framework would supply the guarantee.

5.3 Tier 3: bridges and frontier extensions

G7. The causal-inference and proximal bridge. A singleton is a perfect anchor, and “an anchor that pins a latent quantity” is close to negative controls and proximal causal inference (Miao, Tchetgen Tchetgen) and to instrumental identification. The missing paper recasts the singleton as a proximal or anchor variable and imports the proximal toolkit, generalizing restoration beyond the exact-pin case to noisy anchors. This also connects coarsening to a large, active causal-inference literature the program currently does not speak to.

G8. A learning-theoretic rate. The margin γ and the r/γ^2 rate look like margin-based PAC sample complexity. The missing paper gives a Rademacher or VC treatment of the coarsening hypothesis class, generalizing the rate beyond the stylized linear confounded-subspace model and connecting to the coarse-label learnability line (Fotakis et al. 2021; the 2026 mean-estimation-from-coarse-data work).

G9. Multiclass and longitudinal coarsening. Several instances are binary (weak supervision, MIL); the reliability foundation is time-to-event but the applications are static. Two missing papers: a general multiclass or polytomous coarsening treatment (the weak-supervision paper flags the multiclass extension as open), and coarsening at random for longitudinal or streaming data (coarsened time series, time-varying dropout), which reconnects the program to its survival-analysis origin.

5.4 Cross-cutting gaps

G10. A unified empirical study. Honest limitation: the applications are mostly analytic plus simulation; only scRNA-seq carries real-data drivers. The missing paper runs the rank diagnostic and the singleton-design recipe across several real datasets in different domains, turning the framework’s diagnostics into demonstrated practice. This is the paper that answers a reviewer who says “show me it matters on data.”

G11. The impossibility and limits paper. The program is about when restoration *works*. It should also characterize when it cannot: where no singleton helps (DP, where the singleton defeats the purpose), where the rank deficit is structural and irreducible, where C2 violation is unbounded and no finite external data restores identification. A clean negative-results paper would sharpen the positive ones by drawing their boundary.

5.5 The meta-gap

The most consequential missing artifact may be this monograph’s mature form: a single review or monograph that subsumes the six application notes into worked instances of two or three substantial papers. Submitted as a swarm of short, tightly-coupled, single-author preprints, the program invites a “least publishable unit” reading. Told as the foundation, the two pillars, and a consolidated applications chapter, it reads as what it is: one idea with reach. The gap list above doubles as the table of contents for the chapters such a consolidation would still be missing.

6 How the pieces fit

A reading order for someone entering cold:

1. **masked-causes-in-series-systems** for the C1-C2-C3 collapse and the rank theorem in their original, concrete reliability setting.
2. **coarsening-synthesis** for the abstraction: the one consistency identity, the rank condition, the singleton, and the six corollaries.
3. One application matched to the reader’s field (scRNA-seq is the most developed and the only one with real data).
4. **coarsening-sensitivity** for what happens when C2 fails: the tilt bias bound and the r/γ^2 restoration rate.
5. **mdrelax** for the worked C2-relaxation in reliability, with software.

The dependency spine is: foundation, then synthesis (assumes C2) and sensitivity (relaxes C2) in parallel, then applications instantiate both. The gaps attach as follows: G1, G2, G4, G5, G6 deepen the two pillars; G3 and G10 make them operational; G7, G8, G9 extend their reach; G11 bounds them.

7 Coda: the strategic read

A family of correct, closely-related, single-author papers submitted to selective venues in a short window is vulnerable for reasons that have nothing to do with whether the mathematics is right (it is): fit, perceived significance, and the appearance of slicing one idea thin. The durable assets are intact, every result is a published, citable preprint, and the ideas are timestamped and out in the world. The recoverable move is to change the *unit*: fewer, deeper papers, and a monograph or review as the flagship. The gap list is the agenda for what those deeper papers would contain. The program is not a dead end; it is a spine with several vertebrae still to grow, and it has been told, so far, in too many small voices instead of one.